genfleet

GUC German University in Cairo الجامعة الألمانية بالقاهرة

Kestrel · Milestone 1: Sources, Containers, Cleaning

Data Engineering · Winter 2026 · Semester project · 30% · Due Monday 19 October 2026, 11:59 PM

Covers Lab 1 (sources and storage), Lab 2 (Docker and Compose), Lab 3 (pandas EDA and cleaning).

Read the **Project Overview** first: the scenario, the data, the closed rejection enum, and the semantics you must not guess all apply here. Field detail is in the **Dataset Description**.

Goal

Turn nine files in four formats and three clocks into **one canonical activity table** you can trust, and document every decision you made on the way. This milestone builds pipeline stages 1 and 2: **ingest and normalise**, then **clean and integrate**.

Files you use

FILE	ROLE IN THIS MILESTONE
`agent_activity.parquet`	the main input: every session, LLM turn, and tool call
`tenants.csv`, `agents.csv`, `tools.csv`	reference data that the activity points at
`model_pricing.csv`	prices for the cost metric in Milestone 2; profile and clean it now
`sandbox_policies.json`	the rules each agent should obey; flatten it
`network_flows.csv`	egress attempts
`sandbox_audit.log`	the sandbox's own decisions, as text
`siem_alerts.json`	what the targeted systems saw; link it to the flows

`stream_events.json` and `incident_labels.csv` are not used until Milestone 2.

Tasks

1. Read and profile

Read every batch file, including the ones that are not tables: Parquet read with its schema checked, nested JSON flattened (policies and SIEM alerts), syslog parsed with a regex. Profile what you find (row counts, null rates, cardinalities, distributions, timestamp ranges) and say what is wrong before fixing anything. **Nobody will tell you what is wrong with the data**: the problems you find and how you found them are graded.

2. Clean and integrate

As reusable functions rather than notebook cells:

- normalise every timestamp to UTC, and record how you established each source's convention;
- deduplicate on declared natural keys;

- resolve entities that appear under more than one identifier;
- repair or reject invalid values, and check referential integrity;
- implement the SIEM ↔ network-flow correlation and report matched, ambiguous, and unmatched counts.

Emit **one canonical activity table** (section 5 of the Dataset Description) plus a **rejects table** where every record carries one enum reason and a pointer to its source file and line.

3. Containers

Bring up a Compose stack (PostgreSQL and Adminer suffice) and make the milestone reproducible from a clean clone.

4. First metrics

In pandas, over your curated output, the four Milestone 1 metrics of the overview (numbers 1, 2, 8, and 9): session volume, success and error rate; p50 / p95 session duration; trace completeness and correlation coverage; and the rejected-record and late-event rate. Report each value with the policy behind it. Marking compares your values with reference values computed by the course staff, allowing for differences you have documented.

Deliverables

Commit everything below to the `M1/` folder of your team repository (see section 6 of the overview).

- Profiling notebook
- Reusable cleaning module
- `docker-compose.yml` and run instructions that work from a clean clone
- Curated canonical activity table and rejects table
- **Data dictionary** stating your null, timezone, entity-resolution, dedup, and correlation policies
- **Data-quality findings**: each problem you found, how you detected it, and what you decided
- The four Milestone 1 metrics, each with the policy behind it

Grading scheme (30 marks)

COMPONENT	MARKS	WHAT EARNS FULL MARKS
Reading and profiling	5	Every batch file read in its own format; the profile says what is wrong before anything is fixed
Data dictionary and data-quality findings	5	Each problem found, how it was detected, and the decision taken; null, timezone, entity-resolution, dedup, and correlation policies stated and justified
Cleaning and integration	10	All timestamps in UTC; deduplication on declared keys; entities resolved; one canonical activity table; every rejected record has one enum reason and points back to its source file and line; SIEM ↔ network-flow correlation with a declared rule and tolerance, and matched, ambiguous, and unmatched counts reported
First metrics	8	The four Milestone 1 metrics computed in pandas and close to the reference values, or the difference explained by a documented policy
Engineering and reproducibility	2	Reusable cleaning module rather than notebook cells; `docker compose up` and the run instructions work from a clean clone
Total	**30**	